Reg. No.															

B.Tech. / M.Tech. (Integrated) DEGREE EXAMINATION, NOVEMBER 2025

First Semester

21CYB101J - CHEMISTRY

(For the candidates admitted during the academic year 2025 - 2026)

Note:

i. **Part - A** should be answered in OMR sheet within first 40 minutes and OMR sheet should be handed over to hall invigilator at the end of 40th minute.

ii. **Part - B** and **Part - C** should be answered in answer booklet.

Time: 3 hours **Max. Marks: 75**

PART - A (20 × 1 = 20 Marks) — Answer **all** Questions

		Marks	CO	BL
1.	$K_4[Fe(CN)_6]$ is an example of (A) Tetrahedral (B) Octahedral (C) Square Planar (D) Linear	1	1	1
2.	The crystal field splitting energy for octahedral and tetrahedral complexes is related as (A) $\Delta t \approx 4/9\ \Delta o$ (B) $\Delta t \approx 1/2\ \Delta o$ (C) $\Delta o \approx 2\ \Delta t$ (D) $\Delta o \approx 4/9\ \Delta t$	1	1	2
3.	Which of the following ligands would cause the largest crystal field splitting in an octahedral complex? (A) Cl^- (B) CN^- (C) H_2O (D) NH_3	1	1	1
4.	The shielding constant experienced by a 2p electron in nitrogen atom is (A) 3.45 (B) 3.10 (C) 3.4 (D) 4.7	1	1	2
5.	the chemical often used in salt bridge is (A) KCl (B) KBr (C) $KMnO_4$ (D) KOH	1	1	1
6.	Which condition favors spontaneity in a reaction at constant temperature and pressure? (A) $\Delta G > 0$ (B) $\Delta G = 0$ (C) $\Delta G < 0$ (D) $\Delta H > 0$	1	2	2
7.	Decrease in free energy can be given by $-\Delta G =$ _____ (A) nFE (B) n/FE (C) nF/E (D) F/nE	1	2	1
8.	Corrosion between the dissimilar metals is called as_____ (A) Galvanic corrosion (B) Dry corrosion (C) Oxidation corrosion (D) Concentration cell corrosion	1	2	1
9.	Which of the following representation is generally used to describe conformational analysis in organic molecules? (A) Fisher projection (B) Sawhorse formula (C) Newmann projection (D) Flying-wedge formula	1	3	2

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10. which is equivalent to the identity operation (E) (A) C_1 (B) C_2 (C) C_3 (D) C_4	1	3	2
11. The isomers which can be inter converted through rotation around a single bond are: (A) conformers (B) diastereomers (C) enantiomers (D) positionalisomers	1	3	1
12. Which step in S_N1 reaction is a slow rate determining step? (A) Attack of nucleophile (B) Formation of racemic mixture (C) Formation of carbocation (D) Both a and b	1	3	2
13. Which of the following is not a natural polymer? (A) Rayon (B) Starch (C) Cellulose (D) RNA	1	4	1
14. The random orientation of polymeric chain in a polymer is called (A) Isotactic (B) Atactic (C) Syndiotactic (D) Elastomer	1	4	2
15. The strength of the polymer increases with _____ in molecular weight. (A) Increase (B) Decrease (C) No change (D) Slightly decrease	1	4	2
16. The advantage of using conducting polymers in place of metals is their _____ (A) Cost (B) Light-weight (C) Thermal conductivity (D) Solubility	1	4	2
17. After the proportionality limit in the stress strain curve, we observe _____ (A) Ultimate point (B) Lower yield point (C) Upper yield point (D) Elastic limit	1	5	1
18. The classification of fibers having thin crystals is known as _____ (A) Whisker (B) Fiber (C) Wires (D) Matrix	1	5	1
19. X-ray diffractometers are not used to identify the physical properties of which of the following? (A) Metals (B) Liquids (C) Polymeric materials (D) Solids	1	5	1
20. For constructive interference to take place, the path difference between the two waves should be (A) $n\lambda$ (B) $2n\lambda$ (C) $(n + 1/2)\lambda$ (D) $(2n + 1)\lambda$	1	5	1

PART - B (5 × 8 = 40 Marks) — Marks CO BL

Answer **all** Questions

21. a) Draw energy level diagrams to show the splitting of d orbitals in an octahedral and tetrahedral ligand field.	8	1	3
(OR)			
b) Write a note on the geometrical and optical isomerism exhibited by complexes with suitable examples.	8	1	3

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22. a) Prove $-\Delta A = W_{max}$ and $-\Delta G = W_{useful}$	8	2	3
(OR)			
b) Derive Nernst equation and gives its applications.	8	2	3
23. a) Discuss elements of symmetry with suitable example for each.	8	3	3
(OR)			
b) Explain Cahn-Ingold prelog priority rules to determine R/S configuration on a chiral center taking an example.	8	3	3
24. a) Compare addition with condensation polymerization with examples.	8	4	3
(OR)			
b) Explain in detail n and p doping in conducting polymers.	8	4	3
25. a) Draw and explain all the regions in the stress-strain plot.	8	5	3
(OR)			
b) Compare the roles of matrix and reinforcement in composite materials.	8	5	3

PART - C (1 × 15 = 15 Marks)

Answer **any 1** Question

	Marks	CO	BL
26. a) Draw and explain the potential diagram and discuss in detail the conformational analysis of n-butane.	10	3	4
b) What is the reaction of the following with Cyclopropane? i. Halogens ii. HI iii. sulphuric acid iv. Hydrogen	5	3	2
27. a) Explain Bragg's law with a neat diagram.	5	5	4
b) Discuss the principle, instrumentation and applications of XPS.	10	5	4

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